

Recommended Assessment

Weather Sensor

Measuring Weather Signals

1. What are the units of measurement from the sensor for **Temperature**, **Pressure** and **Humidity**?
2. What did you observe when blowing air into the weather sensor? How do these signals change? Do all of them return to their previous measurements at the same time? If not, which one is the slowest and the fastest?
3. When bringing the device higher or lower, how do the signals change? Are they affected by how fast you move the device? Are they changing consistently with your movement?

Measuring Altitude

4. Considering that the sensor reading gives pressure in Pascals (Pa), how could you get altitude in both feet and meters based on the pressure altitude formula in the lab?
5. What elevation and atmospheric pressure does your location have? How does that compare with the pressure and altitude values from your code?
6. From your altitude plot, what did you observe when bringing your device higher towards the ceiling or lower to the ground? Would you use this device and reading for an absolute or relative altitude change? In what situations would you use this?
7. Based on the operating range for the pressure, what would be the altitudes at which this sensor works?
8. Report the response time under the datasheet parameters for the humidity sensor. How does this value relate to your observed response during the first part of the experiment?
9. Based on section 1 of the datasheet starting in page 8 (1. Specification), what are the resolutions of each of the sensors? What does the resolution tell us about the sensor?

Weather Sensor Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	